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Note

Empirical relationships for remote sensing reflectance and *Noctiluca scintillans* cell density in the northeastern Arabian Sea

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ABSTRACT

A dinoflagellate under the ambit of Harmful Algal Blooms (HAB), the bioluminescent *Noctiluca scintillans* (*NS*), has been infesting the northern Arabian Sea increasingly over the last few decades during late winter. Their occurrence is found to be due to seasonal oscillations in the coastal currents. The physical and biogeochemical parameters associated with the seasonal blooms are reasonably well known. But accurate quantitative estimation capability using remote sensing sensors over the extensive oceanic regime is still lacking. This is especially due to a lack of information on bio-optical properties associated with cell density measurements. We attempted to show that remote sensing reflectance and chl-*a* show significant relationship e.g., $R_{rs}(531)/R_{rs}(510) = 0.8261 + 6.06 \times 10^{-6} NS + 0.02323$chl-*a* ($N = 19$, $R^2_{adj} = 0.99$, $p = 2.5 \times 10^{-17}$, RMSE = 0.1083) which is applicable over diverse areas of the northeastern Arabian Sea e.g., coastal, shelf and offshore regions. The model is supported by a second dataset with an RMSE of 0.022893 ($N = 8$) for the $R_{rs}(531)/R_{rs}(510)$ ratio. The *NS* cell densities were derived from the Rrs(510)/Rrs(531) band ratio within reasonable error and accuracy limits. Including sensor capability at 510 nm is suggested in future satellite launches.

1. Introduction

The northern Arabian Sea is unique in multiple oceanic phenomena. A highly productive region due to intense upwelling in summer, convective mixing in winter and home to extensive fishery resource year round, a strongly suboxic OMZ extending upwards up to 100 m below sea surface, an intensely denitrifying intermediate depth zone etc. make it a natural laboratory (Luis and Kawamura, 2004; SenGupta and Naqvi, 1984). Recent intensification in algal blooms during late winter some of which are harmful, are reported in the past decade (Lotliker et al., 2018; Gomes et al., 2014; Padmakumar et al., 2012). *Noctiluca scintillans* (*NS* hereafter) previously referred as *N. miliaris* is a bioluminescent dinoflagellate, the blooms of which have become a cause of public concern, as it was reported to be caused by untreated sewage from the neighbouring countries and having significant negative impact on fishery resources. From the biogeochemical point of view, there are 2 schools of thought on occurrence and recent intensification of *NS* namely (i) surface suboxicity, due to mixing with sub-oxic intermediate water, and input of anthropogenic organic inputs caused by discharge of untreated sewage (Gomes et al., 2014), and (ii) weakening of winter convective mixing due to north-ward advancing western India Coastal current (WICC) along the eastern Arabian Sea coast (Shankar et al., 2015) leading to weakening of vertical mixing and ceasure of silicate supply and surface suboxicity (Sarma et al., 2019 and references therein; Lotliker et al., 2018; Dwivedi et al., 2012).

A problem with the *NS* remote sensing is in its evolution into a bloom. It is autotrophic due to a photosynthesizing endosymbiont. In the tropical Arabian Sea where it is green (*NS*G), the endosymbiont is *Protoeuglena noctilucae*, a green (prymnesiophyte) alga of the family Marsupiomonadaceae and Order Marsupiomonadales (Saito et al., 2006). Being a dinoflagellate, it is also phagotrophic. The latter property is a primary driver and the availability of prey in the form of a diatom bloom (chl-*a*) acts as a trigger for the *NS* to proliferate in the early stages. This happens at the time of late winter when the sea surface temperature (SST) warms up to 25 °C but not earlier due to the cooler SST. As the diatoms (chl-*a*) disappear by being preyed and silicate loss, the host

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turns to the endosymbiont which works more on nitrate for its photosynthesis as the environment is phosphate-replete (Kumar et al., 2017). The excessive dependence on the endosymbiont, in later stages of the life cycle threatens its own existence (Harrison et al., 2011; Saito et al., 2006). Thus, the *NS* quantification via chl-*a* is likely to be infructuous. The *NS* estimations reported in the NEAS on the basis of Chl-a (Gokul et al., 2019; Gomes et al., 2014; Prakash et al., 2017) may need revisit.

A possible solution to this problem is by directly relating Rrs with *NS* cell density, instead of chl-*a* which shall have a double benefit because chl-*a* was not a dependable predictor of the complex assemblage of phytoplankton biomass even when *NS* were not part of the biome (IOCCG, 2014). This can be made possible if the reflectance pattern of *NS*, in whatever mixed assemblage/suspended matter it is present, is retained as a unique signature. The Rrs spectra of *NS*G exhibit a peak at ~531 nm due to low absorption and high reflectance caused by the green endosymbiont (Gokul and Shanmugam, 2016; Dwivedi et al., 2015), a unique feature. The objective of this Communication is to explore remote sensing reflectance bands and their ratios at various channel bands of the common satellites e.g., MODIS-aqua and SeaWiFS to directly derive *NS* cell densities from key bio-optical parameters.

2. Study area, sampling and analysis

The present study was carried out in the offshore waters of the NEAS (Fig. 1) between 18°N–22°N and 64°E–70°E. In situ measurements were carried out onboard the Indian Ministry of Earth Sciences (MoES) research vessel "Fishery Oceanographic Research Vessel (FORV) *Sagar Sampada*" (cruise no. 356) during the winter of 2017 (6th to 24th February 2017) at 5 stations, two of which were occupied only once and at the other three, time-series sampling at 4 or 6 h intervals and simultaneous onboard observations and analyses were made.

Seawater samples collected from surface (−1 m, a Niskin rosette) were filtered through 47 mm dia Millipore GF/F filters (0.7 μ, nominal) and the filter pads were soon after soaked in 90% acetone and kept in a refrigerator (4 °C) overnight for pigment extraction. Chl-*a* was quantified in the centrifugates (4000 rpm) onboard spectrophotometrically as per Grasshoff et al. (1999). Total Suspended Matter (TSM) was estimated with gravimetrically collected (and washed with purified water, 1–2 mL to remove the adhered sea salt) over pre-heated and weighed 0.22 μ millipore polycarbonate membrane filters followed by drying at 60 °C to constant weight (Pandi et al., 2014). For colored dissolved organic matter (CDOM) measurement, the filtrates from the above were equilibrated to room temperature (22 ± 1 °C) when microbubbles disappear and the absorbance spectra measured onboard with aid of a double beam UV–Visible spectrophotometer (Shimadzu 2600) using 10 cm path length Quartz cells (Pandi et al., 2014). The absorption coefficient of CDOM (m^{-1}) was calculated from the equation $a(\lambda) = 2.303\ A(\lambda)\ 10$ (m^{-1}). For the estimation of *NS*, surface water was collected prior to the sample casts in a clean plastic bucket, and to a well-mixed part of it (1 L) in a plastic (nalgene) bottle, 1% Lugol's iodine in 3% buffered formaldehyde solution in plastic (nalgene) bottles was added followed by enumeration using an epi-fluorescence microscope (Nikon Eclipse, E600). The standard statistical analyses e.g., correlation, regression and the calculation of statistical parameters e.g., bias, mean absolute percentage error (MAPE) and root mean square error (RMSE) were done in Excel 2010 sheets using standard formulae.

2.1. In situ optical data

The AOP (apparent optical properties) were measured as described earlier (Baliarsingh et al., 2017) with a free-falling optical profiler (Satlantic™ hyperspectal radiometer) from 0933 to 1641 h matching

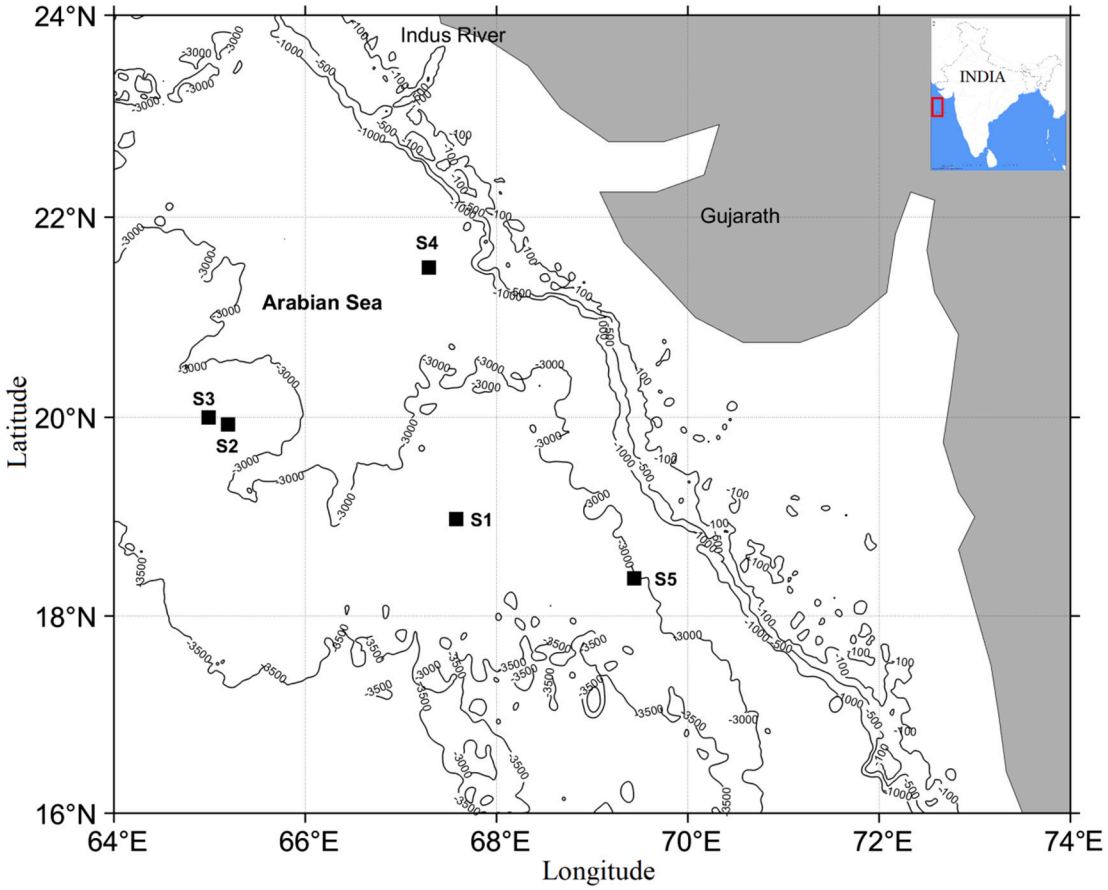


Fig. 1. Study area and station locations (S1 to S5). Inset shows the study region in the NEAS (bathymetry, in meters, is from the high resolution ETOPO1 data).

with equatorial crossing time of MODIS (1330 h). The casts were made 10 m away from the boat to avoid a possible influence of ship shadow. The upwelling/downwelling radiance/irradiance was recorded in the 400–700 nm spectral range. SatView™ software was used for data logging and Prosoft™ 7.7.2 for raw data processing to different data levels. The spectral Rrs(λ) was computed using the equation

$$Rrs(0^+,\lambda) = \frac{Lw\ (0^+,\lambda)}{Ed\ (0^+,\lambda)}$$

where *Lw* (0^+,λ) is the water leaving radiance ($Wm^{-2}nmsr^{-1}$) and *Ed* (0^+,λ) is the downwelling irradiance ($Wm^{-2}nmsr^{-1}$).

3. Results

3.1. Noctiluca scintillans abundance

The cell count was nil at S1, S2 and S5. At S3, the *NS* started to show up beginning Day-2 and reached diurnal maxima for the following two days with 3560 and 4280 cells/L at 1300 h. At S4, the *NS* was noticed from the beginning of its occupation and in much greater abundance. The 1300 h sample on Day 2 showed an intense bloom of *NS*G with a cell count of 52,600/L. On the preceding and succeeding days also maxima occurred in the 1300 h samples, but were weaker at 5450 and 7200 cells/L respectively (Fig. 2a). Stable environmental conditions prevailed during the study. This was demonstrated by the salinity-temperature distribution diagram, e.g., Fig. S1 (supplementary data) in the case of S4. Even profiles ruled out the possibility of transportation from far as causing changes in the abundance of *NS*.

3.2. Chl-a and TSM

The surface chl-a was highest at the time of peak *NS*G (29.35 μg/L) at near noon (1300 h, Day-2) at S4, followed by 2.77 μg/L 6 h later in the evening (1900 h), and 2.29 μg/L at the time when the station was just about reached on the eve of Day 1. TSM did show maximum abundance at peak bloom time (16.79 mg/L), and its profile appeared to follow the *NS* cell count; however, it was more evenly distributed and varied less widely. TSM correlated ($N = 23$) well with both *NS* cell count ($r = 0.87$, $p<10^{-3}$) and chl-*a* (0.75, $<10^{-3}$). The possibility of a dominating role for the peak cell density sample in the high 'r' values was examined by repeating the correlation analysis without this sample ($N = 22$). Highly significant 'r' values of 0.72 ($p < 10^{-3}$) and 0.50 ($p < 2 \times 10^{-2}$) respectively were still obtained. However with chl-*a*, the corresponding coefficients were 0.92 (N = 23, $p < 10^{-3}$) and 0.27 (N = 22, <0.5), which indeed indicated the dominating role on it of the *NS*-peaking sample.

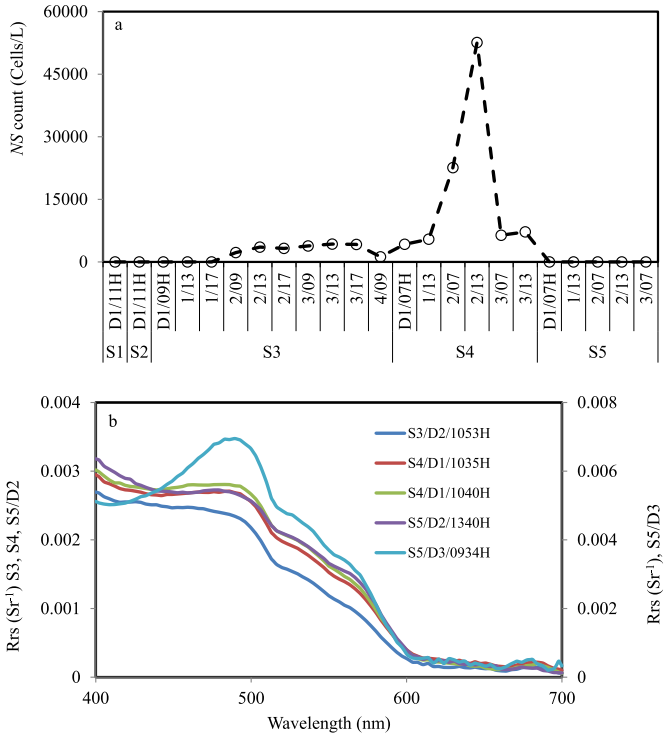


Fig. 2. (a) *Noctiluca scintillans* cell counts on collections during the day (D1/11H… etc. are for Day 1, 1100 h and so on); (b) Representative remote sensing spectra at S3 (2), S4 (2) and S5 (1).

3.3. CDOM optical characteristics

The surface CDOM (a_{440}) was quite low particularly at S4 (range: 0.1152–0.2764, average = 0.1919 m^{-1}) where large *NS*G cell densities were observed compared to S3 (highest 0.4155, average 0.1888) where *NS*G cell count was either low or nil. CDOM did not show correlation and had a scattered distribution versus *NS*, chl-*a* or TSM.

3.4. Remote sensing reflectance spectra

A total of 27 Rrs spectra were acquired one each from S1 and S2, and 6, 12 and 7 from S3, S4 and S5 respectively. The Rrs spectra (Fig. 2b) showed a pattern of gradually decreasing reflectance from 400 nm towards longer λ with a steep fall at ~600 nm and a gentle fall thereafter. The 400–600 nm region which is the focus of this study is populated with a significant peak at ~490 nm and shoulders/broad peaks at ~510, ~530 and ~ 550 nm. In the *NS* peak abundance sample, the ~550 nm was the most prominent and appeared to mask other minor peaks, if any. The Rrs band ratios were calculated for different pairs of wavelengths where inflections and peaks are shown in the Rrs spectra. These are the MODIS-aqua band centres and additionally 510 nm of the SeaWiFS. Focus is centred on ratios involving 443, 488, 510, 531 and 551 nm, as better relationships were noticed for them over the terminal (visible) bands of 412 and 667 nm. The *NS* cell density correlated very significantly with the Rrs band ratio Rrs(531)/Rrs(510) ($r = 0.932$, $N = 27$, $p < 10^{-4}$) (Table 1) followed by Rrs(551)/Rrs(510) (0.917) and Rrs(510)/Rrs(488) (0.911). Also, the Rrs(531)/Rrs(510) band ratio correlated highly significantly with chl-*a* (0.95) (Table 1), the next best ratio being Rr(551)/Rrs(510) (0.945). With TSM, the 'r' values were also significant

Table 1
Pearson correlation coefficient (r) matrix of constituents* (only the values discussed in Text are given).

	NS	Chl	TSM	$R_{rs}(531)/R_{rs}(510)$	$R_{rs}(510)/R_{rs}(531)$
DSM (N = 19)[a]					
Chl	0.9143				
TSM	0.8685	0.7491			
$R_{rs}(531)/R_{rs}(510)$	0.9317	0.9497	0.7794		
$R_{rs}(510)/R_{rs}(531)$	−0.9659	−0.9285	−0.8388	−0.9763	
$R_{rs}(510)/R_{rs}(551)$	−0.9318	−0.8792	−0.8340	−0.9346	0.9912
DSV (N = 8)[a]					
Chl	0.9336				
TSM	0.8785	0.8762			
$R_{rs}(531)/R_{rs}(510)$	0.9864	0.9683	0.9121		
$R_{rs}(510)/R_{rs}(531)$	−0.9801	−0.9213	−0.9150	−0.9884	
$R_{rs}(551)/R_{rs}(510)$	0.9777	0.9669	0.9395	0.9966	−0.9859

[a] DSM = Modelled dataset, DCV=Validating dataset.
* All r values are significant at $p < 10^{-3}$.

($p < 10^{-3}$) but lower (0.779 and 0.767 respectively). Thus, although there may be more bio-optical constituents showing correlation to Rrs (531)/Rrs(510), *NS* abundance and chl-a were the key constituents from which Rrs(531)/Rrs(510) could be predicted.

3.5. Rrs models

Multiple linear regression analysis was attempted for modelling the Rrs(531)/Rrs(510) band ratio in terms of *NS* cell density and chl-*a*. As the spectral radiometer and Niskin rosette casts were not simultaneous, the data of each spectral radiometer (RM) deployment was connected with the biogeochmical data of the sample that was collected nearest to it in time (model dataset, DSM, $N = 19$). A second dataset was of the left out Rrs spectra that matched with the same water sample(s) but with greater time gap (validation dataset, DSV, $N = 8$). The time lag of sample collection from the RM operation varied within ±3 h for both DSM (170 to −86 min) and DSV (163 to −145). This was necessitated because of the shorter time window available for the operation of the RM.

The DSM yielded Eq. (1) which is highly significant ($p < 10^{-4}$). The t-stat (125.2, 5.2, 12.9 respectively) and p values ($<10^{-4}$) of the individual coefficients in Eq. (1) are also of high significance.

$$R_{rs}(531)/R_{rs}(510) = 0.8261 + 6.06 \times 10^{-6}\mathrm{NS} + 0.02323\mathrm{chl} - a\ (N = 19, R^2_{adj} = 0.99, p < 10^{-4}, \mathrm{RMSE} = 0.1083)\ldots \quad (1)$$

The RMSE, MAPE and bias statistics of the estimated band ratio improved significantly without the *NS* peak-cell count datum (Table 2). The Rrs band ratio calculated from Eq. (1) deviated between −4.4 and 30.2% from the observed values. The distribution of the model estimate of $R_{rs}(531)/R_{rs}(510)$ versus the observed band ratio was linear and highly significant at $p < 10^{-3}$ (Fig. 3a). This equation was then applied to compute the $R_{rs}(531)/R_{rs}(510)$ band ratio from *NS* cell density and chl-*a* on the DSV ($N = 8$). A good RMSE (0.02234) for the $R_{rs}(531)/R_{rs}(510)$ ratio, and % deviation of the calculated from the observed band ratio (−4.5 to 3.6%) supported the model.

Among several other ratios attempted, the $R_{rs}551/R_{rs}510$ band ratio gave a nearly significant model (Eq. (2)) in which the RMSE was 0.03831 and the prediction error ranged from −6.6 to 14.2%. When this model was applied on the DSV, the RMSE and the lowest

$$R_{rs}(551)/R_{rs}(510) = 0.6633 + 8.28 \times 10^{-6}\mathrm{NS} + 0.03855\mathrm{chl} - a\ (N = 19, R^2_{adj} = 0.99, p < 10^{-4})\ldots \quad (2)$$

negative error further improved (0.2830, −3.3%); however, the positive error was higher (46.1%).

Table 2
Statistical performance of the models.

Dataset	N	*NS* range	Modelled property (model)	RMSE	MAPE (%)	Bias
DSM	19	0 to 52,600	$R_{rs}(531)/R_{rs}(510)$ (Eq. (1))	0.1083	3.9	0.02435
DSM	18	0 to 22,600	-do-	0.024133	2.4	0.00010
DSM	19	0 to 52,600	*NS* (Eq. (3))	3639	42.1*	1873
DSM	18	0 to 22,600	-do-	3737	48.0#	1952
DSM	8	2240 to 52,600	-do-	3274	42.1	0
DSV_1	36	0 to 5500	-do-	2437	59.1*	1231

For *NS*-present samples: *N = 8, #*N* = 7.

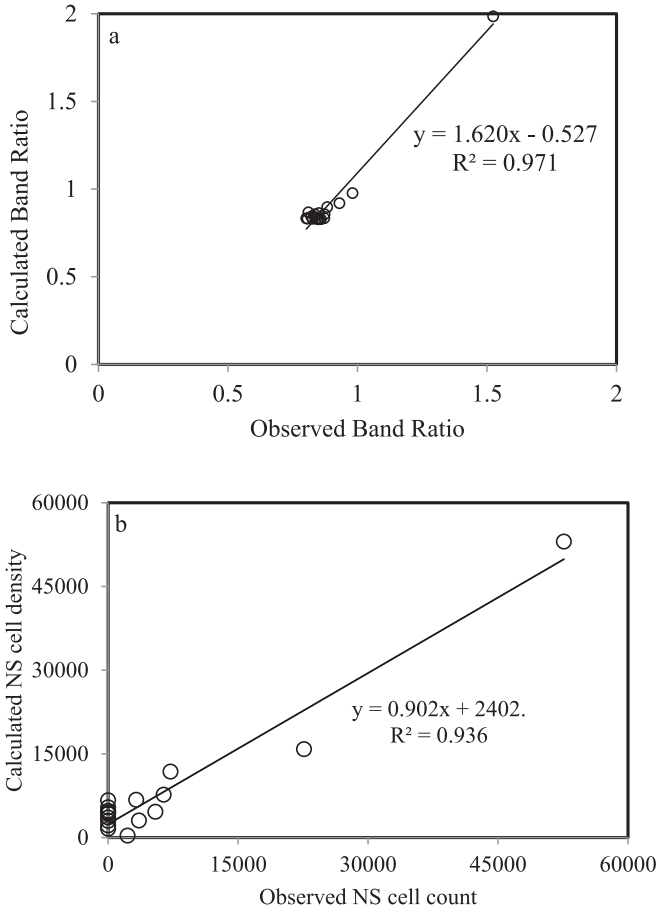


Fig. 3. Model performance with DSM dataset: (a) Rrs band ratio ($R_{rs}(531)/R_{rs}(510)$ ratio) calculated from Eq. (1) versus the observed ratio, (b) *NS* cell density calculated from Eq. (3) versus the actual cell count. The RMSE, MAPE and bias of these estimates are given in Table 2.

3.6. NS models

The DSM had 11 samples (out of the 19) in which the *NS* cell density was nil and the DSV none (0/8). As the large number of *NS*-nil samples in DSM (11/19) may unduly bias the overall model, and because DSV had none of *NS*-nil samples, attempt was made to fit cell density of only *NS*-present samples (of DSM) in terms of the Rrs band ratio, and then to examine the model performance on extrapolation to zero *NS*. Of all the ratios, the Rrs(510)/Rrs(531) band ratio gave the best and consistent relationship. Including other dependent variables like chl-*a* or TSM in multiple regression decreased the significance.

$$NS\ \mathrm{cells/L} = 89,360 - 72,133 \times \left((\mathrm{Rrs}(510)/\mathrm{Rrs}(531)\right), R^2 = 0.96, N = 8, \mathrm{RMSE} = 3274, p < 10^{-3} \quad (3)$$

The prediction error (RMSE) of Eq. (3) is 3274 (Table 2) and 4664 *NS* cells/L in DSM and DSV respectively. When this model was extrapolated to cover *NS* nil samples of the total dataset (N = 19), the statistics did not suffer much (Table 2). The scatter of the model estimates of *NS* cell density versus the observed cell count (Fig. 3b) retained linearity even with *NS* nil samples included; however, their exclusion produced nil bias and a MAPE of 42.1% for the *NS* estimate (Table 2).

4. Discussion

Modelling the in situ measured Rrs in terms of the optical parameters like phytoplankton biomass and its composition, suspended inorganic particles, DOM etc. is an essential requirement in HAB tracking. The

other requirement is a match-up with increasing confidence of the modelled reflectance for the satellite records retrieved and corrected (Blondeau-Patissier et al., 2014). Remote sensing satellites can measure bio-optical components with increased resolution over the years. This is when the algorithms have incorporated the phytoplankton (especially HAB) species/group-specific reflectance properties (Zhu et al., 2019).

In the case of *NS*, the reflectance behaviour in the field is complicated from its unique attributes namely (i) its group/species specific carotenoid composition showing variable spectral absorption/reflectance properties especially in the 480–580 nm region; *NS* is richer in carotenoids than in chlorophyll (Van Mol et al., 2007) (ii) its large cell size ranging from 0.2 to 2 mm (average, 0.5 mm) compared to other phytoplankton (Guiry and Guiry, 2020), (iii) it's buoyancy due to ammonia producing vacuoles because of which it is enriched at surface, (iv) its gelatinous/slimy nature because of which it flocs into mats that produce floating patches, (v) the migrating patches that follow wind, currents and food, and (vi) the endosymbiont which imparts green (or red) colour, more prominently in the later stage of its life cycle when diatom food on which it fed earlier is no longer available. This is perhaps why only a limited progress is made in *NS*G bloom detection (Baliarsingh et al., 2017).

In the green-yellow leaves of plants, the reflectance around 520 nm correlated closely with carotenoids content and used as a measure of carotenoids content in them (Zur et al., 2000). In the present study, Rrs (510) emerges as a key input for computing NS cell density.

4.1. The Rrs(531)/Rrs(510) ratio

In remote sensing algorithms, the band ratios considered are usually with Rrs(551) or Rrs(555) since many Rrs spectra have maxima and all pigments (chl-*a*, carotenoids) have least absorption in the 550–570 nm region. The scattering by non-organic suspended matter and phytoplankton cell walls also occurs in this region (Blondeau-Patissier et al., 2014). Since the Rrs (and *NS* cell density) showed good correlations with TSM, attempting ratios other than with Rrs(551) is likely justified. Further, as the reflectance (peak) of *NS*G at 531 (Gokul and Shanmugam, 2016) would also increase the neighbouring 551 nm reflectance, for a positive relation to expect, the Rrs(531) needs be in the numerator. Similarly, the λ where reflectance is low (e.g., 510 nm), if used as the normalizer (denominator), would be accentuating the likely positive influence of 531 nm, besides cancelling the measurement errors. In fact, different combinations of ratios were attempted, and all were found to be inferior for predicting $R_{rs}(531)/R_{rs}(510)$ ratio from the *NS* cell count and chl-*a*. The ratio that was next nearest in terms of performance is the Rrs(551)/Rrs(510) ratio. For accurate models Rrs(510) is preferred. So far, only the 443, 488, and 531 bands have been in currency for *NS*G and the red λs of 667 and 668 in the case of *NS*R (Baliarsingh et al., 2017). As an alternative to Rrs(510), Rrs at other λ were also attempted, but not presented here. All those band ratios e.g., Rrs(531)/Rrs(488) yielded equations which performed with less rigorous statistics. Hence, future launches e.g., OCM-III (Gunter's space page, n.d.) could consider 510 nm channel capability.

4.2. Key role of NS and chl-a in Rrs

The Eqs. (1) and (2) are highly significant and are applicable over the entire range of *NS* cell densities and chl-*a* measured. The positive relationships (Eqs. (1) and (2)) indicate that as the cell density and chl-*a* increased, the Rrs(531) increased and/or that the Rrs(510) decreased, supporting the hypothesis. In most reports, the *NS* count is either less or on a similar scale (Sarma et al., 2019 and references cited therein). A lone report of the cell density surpassing the presently observed has come to our notice (Madhu et al., 2012) south of the NEAS region. The Rrs band ratio can be predicted within reasonable error limits. *NS* and chl-*a* appear to exert decisive role, subsuming the role of other optical constituents e.g. inorganic suspended particles, detritus and CDOM in the Rrs spectra. This is especially so in the case of S4 and S3 (partly) where *NS* was observed. S5 being a coastal station and because S1, S2 and S5 stations have a different phytoplankton (chl-*a*) and particle distribution, they differed from the *NS*-prone waters of the NEAS (Sarma et al., 2019).

4.3. NS cell density

Eq. (3) is also of high significance. Within the oceanic regions to which *NS* is endemic (Harrison et al., 2011), it can potentially be applied reliably. Such regions are well characterized (Lotliker et al., 2018, 2020; Sarma et al., 2019; Dwivedi et al., 2016). The Eq. (3) cannot directly distinguish with a high degree of confidence regions having little/nil *NS* from regions in which *NS* cells are present as in those samples ($N = 11$), the RMSE was 3883 and a predicted maximum cell count upto ~6800 cells/L with $NS = 0$ (observed) samples. However, all $NS < 5000$/L samples had TSM < 7 mg/L, and $NS > 5000$/L samples had TSM > 7 mg/L (Supplementary Data). Thus, the Eq. (3) may be applied with little errors in prediction when *NS* cell densities>6800 and TSM > 7 mg/L.

We attempted to verify its utility on an unknown dataset (unpublished, designated as DSV_1) acquired in the NEAS during 4th to 14th February 2019 onboard FORV Sagar Sampada (Cruise No. 383). Eleven samples were collected once each day, and 36 RM casts were deployed within the ±3 h window. The *NS* cell counts were nil on 9 days, and the other 2 days had 5500 (Day 7) and 3400/L (10) (Supplementary Data). The DSV_1 although dominated by nil *NS* abundance samples, agreed satisfactorily with the *NS* model with bias, RMSE and MAPE of 1231, 2437 and 59.1% respectively (Table 2).

The cache with *NS* is that even when it is observed as a minor constituent of the total phytoplankton e.g., in the coastal Southern California (Kimor, 1979), due to its buoyancy, the concentrations can be considerably higher in the top few centimeters of the water column (Harrison et al., 2011). In this study, to be able to posit our results with surface reflectance behaviour, we enumerated *NS* at the upper surface. The (bucket) method of collection may compromise on the accuracy of *NS* counts but we observed that the error was limited to ±20%. The *NS* cell counts are considered as more realistically represented by the in situ Rrs spectra i.e., *NS* cell counts computed. More studies are needed to verify whether the models apply for the *NS*G in the northern Indian Ocean in general.

5. Conclusion

Tracking of *Noctiluca scintillans (NS)* is extremely difficult due to its ephemeral occurrence (Qi et al., 2019). Moreover, the (Green) *NS* cannot be differentiated from other phytoplankton solely on remote sensing spectroscopy (Thibodeau et al., 2014). Since the blooms are sequel to a sequence of connected oceanographic phenomena such as which occur in the NEAS, in specified areas, models should be expected to work best if they take the sea truth into greater account, as shown in here. The Eqs. (1) to (3) are linear since suitable Rrs ratios were used, with ease of application. Eqs. (1) and (2) are associated with high degrees of accuracy in predicting the Rrs(531)/Rrs(510) and Rrs(551)/Rrs (510) respectively. In the case of Eq. (3), an initial remote mapping of total suspended matter can potentially be considered that may narrow down analysis to appropriate samples (regions) for estimating *NS* cell densities. More observations in diverse ecosystems and a refinement of the conclusions shall make case for including 510 nm reflectance sensing capability in future launches of the Earth orbiting platforms for phytoplankton blooms' e.g., *NS* tracking and quantification and so that the high resolution mapping would in turn help resolve the possible role of anthropogenic effects on the expanding blooms.

CRediT authorship contribution statement

The author contributions were as follows for this paper:
SRP: Onboard observations and analysis, Laboratory Analysis,

Manuscript preparation.

SKB: Onboard observations and analysis, Laboratory Analysis, Manuscript.

AAL: Conceptualization, Onboard observations and analysis, Laboratory Analysis.

NSS: Data processing, Manuscript writing.

SCT: Laboratory methods, Technical support.

Declaration of competing interest

Authors declare no competing financial or other interest.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.marpolbul.2020.111770.

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